Sahaj Bhadja

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Summary

Software engineer with 2+ years building **tested systems software**, simulation-style tools, APIs, and CI/CD-backed workflows; strong in C/C++, Python, Linux, embedded/RTOS concepts, fault-tolerant design, verification, and documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **fault-tolerant platform workflows** for 500+ users, translating requirements into APIs, validation jobs, background execution paths, release tasks, and operational documentation.
- Improved **runtime reliability and performance** by tuning indexed queries, asynchronous jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms**.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **Python and JavaScript services** handling 10,000+ daily requests, implementing REST APIs, concurrent file processing, validation checks, and production bug fixes.
- Supported **integration and verification cycles** by reproducing OCR, persistence, and execution defects, documenting root causes and reducing feedback turnaround by **60%**.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Created **Python simulation and evaluation pipelines** across 100,000+ records, producing reproducible metrics, logs, and reports for model validation and debugging.
- Documented **technical processes and experiment behavior** for researchers, turning ad hoc scripts into repeatable workflows with clearer handoffs and traceable outputs.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **cross-platform microservices and ETL workflows** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting data movement.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito across validation flows, increasing CI reliability by **68%** and reducing regressions.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **modular service features** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across distributed workflows.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** and improving repeatable deployments and troubleshooting documentation.

Skills

Systems Languages: C, C++, Python, Java, Rust familiarity, SQL, Bash, JavaScript/TypeScript
Embedded & Safety: RTOS concepts, embedded Linux familiarity, fault tolerance, fault mitigation, safety-critical process awareness
Testing: Unit tests, integration tests, PyTest, Mockito, automated testing, verification support, data collection, reports
Simulation & Systems: Process scheduling, concurrency, memory management, Linux/Unix, modeling pipelines, cyber-physical systems interest
Delivery: Git/GitHub, CI/CD, Jenkins, Docker, code reviews, Agile/Scrum, requirements documentation, technical design notes

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **C++ scheduling algorithms** including FCFS, SJF, SRT, RR, priority, and MLQ with metric calculation and visualization across 1K+ process simulations.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a **containerized REST processing API** with AWS deployment, CI/CD releases, validation checks, and reproducible execution, achieving **87%** classification accuracy.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and database management through Java services and TypeScript interfaces.